

# Haoxue Ran

Department of Environmental and Municipal Engineering

Xi'an University of Architecture and Technology

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## EDUCATION

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**Xi'an University of Architecture and Technology**

**Sep 2021 – Expected Jun 2027**

Ph.D., Environmental Engineering (*Direct-Track, “Cuiying” Excellence Program*)

Advisor: Prof. Rui Miao

**Guilin University of Technology**

**Sep 2017 – Jun 2021**

B.E., Environmental Engineering (*Elite Engineering Program*)

Academic Rank: 1/30 within the program

## RESEARCH EXPERIENCE

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**Graduate Student Researcher**

**Xi'an University of Architecture and Technology**

Advisor: Prof. Rui Miao

**2022 – Present**

Department of Environmental and Municipal Engineering

- Developed a trace-dose Fenton pretreatment technology at neutral pH to mitigate membrane organic fouling, and elucidated its underlying mechanisms.
- Established a systematic analytical framework based on data-driven models to predict membrane fouling behavior and identify its key governing factors.
- Quantitatively decoupled the contributions of individual causal pathways to membrane fouling mitigation by integrating causal inference with micro-interfacial analysis techniques.
- Investigated the different mechanisms of divalent metal ions ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ) on the efficacy of Fenton pretreatment in membrane organic fouling mitigation.
- Quantitatively decoupled the key factors and causal relationships of  $\text{Ca}^{2+}/\text{Mg}^{2+}$  ions governing Fenton pretreatment for membrane fouling mitigation by integrating artificial intelligence with molecular dynamics simulations.

## PUBLICATIONS & PATENTS

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### *Journal Papers*

1. **Ran, H.**, Miao, R.<sup>\*</sup>, Zhao, J.<sup>\*</sup>, Zhang, H., Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Quantitative decoupling of the key factors and mechanisms of Fenton-pretreated membrane fouling mitigation via machine learning-assisted causal inference. *Water Research*, 303, 126247.
2. **Ran, H.**, Wang, Y., Miao, R.<sup>\*</sup>, Zhao, J.<sup>\*</sup>, Wang, S., Zhang, J., Wang, P., Wang, L. and Yin, J. **2026**. Effects of feature selection on predictive performance and interpretability in data-driven modeling:

A case study of membrane flux prediction after Fenton pretreatment. *Separation and Purification Technology*, 393, 137304.

3. Miao, R. \*, **Ran, H.**, Yang, Y., Li, Y., Ma, Z., Lv, Y., Meng, X., He, M. and Wang, L. **2024**. In situ acid production by organic matter induced with trace homogeneous Fenton reagent for membrane fouling control. *Water Research*, 258, 121752.
4. Miao, R. \*, Wang, Y., **Ran, H.**, Li, Y., Yang, C. and Wang, L. **2023**. Relative importance of drag force and adhesion forces in the fouling of ultrafiltration membranes by secondary effluent organic matter after pre-ozonation. *Journal of Membrane Science*, 678, 121677.

#### *Patents & Software Copyrights*

- Miao, R., **Ran, H.**, Wang, L., Lv, Y., Yang, C. and Wang, Y. (**2023**). *Method and application for membrane fouling control via  $\text{Fe}^{2+}$ - $\text{H}_2\text{O}_2$  induced in situ acid production from organic pollutants*. Chinese Invention Patent, Application No: CN202310381227.4.
- Miao, R., **Ran, H.**, Zhao, C., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method for membrane fouling mechanism identification and regulation based on causal inference*. Chinese Invention Patent, Application No: CN202511264800.9.
- Miao, R., **Ran, H.**, Yang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Machine learning-based method and system for the optimal control of membrane fouling pretreatment*. Chinese Invention Patent, Application No: CN202510539314.7.
- Miao, R., **Ran, H.**, Wang, Y., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Method and system for membrane fouling prediction and key factor identification/regulation in Fenton pretreatment*. Chinese Invention Patent, Application No: CN202511401234.1.
- Bai, S., **Ran, H.**, Wu, H. and Zhou, P. (**2020**). *Intelligent maintenance device for green building vegetation*. Chinese Utility Model Patent, Patent No: ZL202020307040.1.
- **Ran, H.**, Miao, R., Wang, S., Lei, Y. and Zhang, J. (**2025**). *Dosage optimization system for Fenton pretreatment-based membrane fouling mitigation under various operating conditions*. Chinese Computer Software Copyright, Registration No: 2025SR2214520.

## **RESEARCH PROJECTS & GRANT PROPOSALS**

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### **National Natural Science Foundation of China (NSFC, General Program)**

*Mechanisms and mitigation strategies of inorganic salt crystallization fouling on ultrafiltration membranes during lithium extraction from salt lake brine.*

Role: Co-drafted proposal.

**2026 (Proposal Under Review)**

### **Natural Science Basic Research Program of Shaanxi Province (Youth Project, Class A)**

*Research on theory and application of membrane fouling control based on multidimensional influencing factors.*

Role: Co-drafted proposal & Graduate Research Assistant.

**2026 – Present**

### **National Key Research and Development Program of China (Sub-project)**

*Interfacial migration mechanisms of salt ions in brine and fouling prevention/control technologies for membranes/adsorbents.*

Role: Graduate Research Assistant.

2022.10 – 2026.09

**National Natural Science Foundation of China (NSFC, General Program)**

*Mechanism and regulation of hydration repulsion for membrane fouling mitigation during ultrafiltration.*

Role: Graduate Research Assistant.

2021.01 – 2024.12

**Youth Innovation Team of Shaanxi Universities**

*Innovative Team of Membrane Separation Technology for Valuable Resource Extraction from Salt Lake Brine.*

Role: Graduate Research Assistant.

2023.01 – 2026.12

**International Science and Technology Cooperation Project of Shaanxi Province**

*High-efficiency reuse process of municipal secondary effluent centered on ultrafiltration technology.*

Role: Graduate Research Assistant.

2022.01 – 2023.12

**National Natural Science Foundation of China (NSFC, General Program)**

*Self-cleaning mechanism and regulation of microorganisms on MBR membrane surface induced by self-forming dynamic membranes.*

Role: Research Contributor.

2024.01 – 2027.12

**Enterprise-funded Collaborative R&D Projects**

*Resource utilization of kitchen and septic tank domestic wastes in Shendong Coal.*

Role: Research Contributor.

2022.07 – 2026.12

*Recycled utilization of mine water for promoting low-carbon emission reduction in boilers.*

Role: Research Contributor.

2025.05 – 2025.11

## **HONORS & AWARDS**

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*Research & Innovation Awards*

- Third Prize, 19<sup>th</sup> “Challenge Cup” National Undergraduate Academic Science and Technology Competition 2025
- Best Presentation Award, 20<sup>th</sup> National Graduate Environmental Forum (Tongji University) 2025
- Outstanding Undergraduate Thesis Award, Guilin University of Technology 2021
- Third Prize, 16<sup>th</sup> “Challenge Cup” National Undergraduate Academic Science and Technology Competition 2019
- Grand Prize, “Challenge Cup” National Undergraduate Academic Science and Technology Competition (*Guangxi Regional Selection*) 2019

*Scholarships & Honors*

- “Yu Dan” Corporate Excellence Scholarship, Xi’an University of Architecture and Technology

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|-------------------------------------------------------------------------------|------|
|                                                                               | 2025 |
| • Second-class Scholarship of Xi'an University of Architecture and Technology | 2025 |
| • Second-class Scholarship of Xi'an University of Architecture and Technology | 2024 |
| • First-class Scholarship of Xi'an University of Architecture and Technology  | 2023 |
| • Second-class Scholarship of Xi'an University of Architecture and Technology | 2022 |
| • Outstanding Graduate Honor, Guangxi Autonomous Region, China                | 2021 |
| • Guangxi Provincial Government Scholarship                                   | 2020 |
| • Special-class Scholarship of Guilin University of Technology                | 2020 |
| • Outstanding Student Award, Guilin University of Technology                  | 2020 |
| • First-class Scholarship of Guilin University of Technology                  | 2019 |
| • Outstanding Student Award, Guilin University of Technology                  | 2019 |
| • First-class Scholarship of Guilin University of Technology                  | 2018 |
| • Elite Engineer Fellowship, Guilin University of Technology                  | 2018 |
| • Outstanding Student Award, Guilin University of Technology                  | 2018 |

## RESEARCH PRESENTATIONS

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1. "Decoupling Fenton-Mitigated Membrane Fouling Mechanisms via Causal Inference and Machine Learning," *8<sup>th</sup> IWA Regional Membrane Technology Conference (IWA-RMTC 2026)*, Shanghai, China, Aug 2026. (Oral Presentation).
2. "Deciphering Complex Membrane Fouling Mechanisms via Explainable Machine Learning: A Case Study of Fenton-Pretreated Membranes," *Academic Exchange on Green Low-Carbon and Sustainable Development (IEEPA Annual Meeting)*, Hangzhou, China, Dec 2025. (Oral Presentation).
3. "Mechanisms of Trace  $\text{Fe}^{2+}/\text{H}_2\text{O}_2$  Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *2025 China Membrane Industry Intelligent Manufacturing and Application Technology Conference*, Xi'an, China, Nov 2025. (Oral Presentation).
4. "Mechanisms of Trace  $\text{Fe}^{2+}/\text{H}_2\text{O}_2$  Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *20<sup>th</sup> National Graduate Environmental Forum (Tongji University)*, Shanghai, China, May 2025. (Oral Presentation, **Best Presentation Award**).
5. "Mechanisms of Trace  $\text{Fe}^{2+}/\text{H}_2\text{O}_2$  Pretreatment on Organic Fouling Behaviors of Ultrafiltration Membranes," *19<sup>th</sup> National Environmental Ph.D. Academic Conference (Tsinghua University)*, Beijing, China, May 2025. (Oral Presentation & **Session Chair**).

## TEACHING AND MENTORING EXPERIENCE

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|---------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Graduate Teaching Assistant</b>                                                    | <b>Xi'an University of Architecture and Technology</b> |
| Course: <i>Membrane Separation Theory and Technology</i> (Instructor: Prof. Rui Miao) | <b>Spring 2022</b>                                     |

- Assisted in course logistics, graded homework assignments, and provided feedback to the instructor (about 120 students).

### Research Seminar Coordinator

Shaanxi Membrane Separation Research Institute

Miao Research Group

2021 – Present

- Scheduled academic topics and managed logistics for monthly large-scale research group seminars (about 20–30 participants).

### Mentoring and Research Advising

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|------------------------------------------|----------------|
| • Jiayi Zhang (Undergraduate Researcher) | 2022 – Present |
| • Yifan Yang (M.S. Student)              | 2022 – 2025    |
| • Siyu Wang (M.S. Student)               | 2023 – 2026    |
| • Yuqing Lei (M.S. Student)              | 2024 – Present |
| • Fangru Zhu (M.S. Student)              | 2025 – Present |

## PROFESSIONAL SERVICE AND LEADERSHIP

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**Assisted Review:** *Water Research, Journal of Membrane Science, Chemical Engineering Journal, Desalination, Separation and Purification Technology, Journal of Water Process Engineering.*

**Student Laboratory Manager:** Miao Research Group, Shaanxi Membrane Separation Research Institute.

**Safety & Operations Coordinator:** Department of Environmental and Municipal Engineering, Xi'an University of Architecture and Technology.

## RESEARCH SKILLS

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**Analytical Characterization:** Liquid Chromatography-Time of Flight Mass Spectrometry (LC-TOF-MS), Quartz Crystal Microbalance with Dissipation (QCM-D), Confocal Laser Scanning Microscopy (CLSM), Liquid Chromatography with Organic Carbon Detection (LC-OCD), Scanning Electron Microscopy (SEM), Inductively Coupled Plasma Mass Spectrometry (ICP-MS), Atomic Force Microscopy (AFM), Fourier Transform Infrared Spectroscopy (FTIR), Zeta Potential & Hydrodynamic Diameter Analysis, Laser Diffraction Particle Size Analysis.

**Modeling & Computation:** Data-driven predictive modeling (CatBoost, TabPFN, Random Forest, Deep Learning), Explainable Artificial Intelligence (XAI), Causal Inference & Directed Acyclic Graphs (DAGs), Molecular Dynamics (MD) Simulations.